

**1 Problem**

- Autoregressive models are slow to sample, while adversarial training can be unstable and omit modes.
- Diffusion offers a likelihood-based route but previously lagged the best image generators.

**2 Forward process**

GRADUALLY DESTROY DATA

- A fixed Markov chain adds small Gaussian noise over  $T=1000$  steps until images approach an isotropic normal distribution.
- The closed-form  $q(x_t|x_0)$  lets training sample any noise level directly rather than simulate all preceding steps.

**3 Reverse model**

DENOISE ONE STEP AT A TIME

- A time-conditioned U-Net predicts the Gaussian noise added to  $x_t$ ; this parameterizes the reverse transition toward  $x_0$ .
- $L_{\text{simple}} = E[t, x_0, \epsilon] || \epsilon - \epsilon_{\theta}(x_t, t) ||^2$

DDPM

**Denoising Diffusion Probabilistic Models**

Ho, Jain &amp; Abbeel | arXiv:2006.11239v2

Learn image generation as the reversal of a fixed Gaussian corruption process, turning score estimation into a stable denoising objective.

diffusion

noise prediction

score matching

**4 Theory & design**

ELBO MEETS SCORE MATCHING

- A weighted variational bound connects diffusion training to denoising score matching and Langevin dynamics.
- Attention at 16x16 resolution, a linear variance schedule and exponential moving averages stabilize synthesis.

**5 Evidence**

STATE OF THE ART CIFAR-10

- Unconditional CIFAR-10 reaches Inception Score 9.46 and FID 3.17; 256x256 LSUN quality is similar to ProgressiveGAN.
- The reverse chain also yields progressive lossy decomposition, linking generation and rate-distortion behavior.

**6 Meaning & limits**

- DDPM showed that a simple denoising loss can produce high-quality, diverse samples and launched modern diffusion models.
- The original sampler needs about 1000 sequential network evaluations, making generation slow and compute intensive.